$P_m(a) = x,\ a \perp m$

$\lambda(m)$

$d(\lambda(m)) \ni x$

Задача: Вычислить $P_{23}(7)$

$\lambda(23) = \varphi(23) = 22$

$d(22) = \{1, 2, 11, 22\}$

$7^2 \overset{23}{\equiv} 49 - 23 \cdot 2 \overset{23}{\equiv} 3 \neq 1$

$7^{11} \overset{23}{\equiv} (7^2)^5 \cdot 7 \overset{23}{\equiv} 3^5 \cdot 7 \overset{23}{\equiv} 3^2 \cdot 27 \cdot 7 \overset{23}{\equiv} 9 \cdot 4 \cdot 7 \overset{23}{\equiv} 9 \cdot 28 \overset{23}{\equiv} 9 \cdot 5 \overset{23}{\equiv} 45 \overset{23}{\equiv} -1$

$\downarrow$

$7^{22} \overset{23}{\equiv} 1$

Ответ: 22

$q \in \mathbb{P}_{\mathbb{N}}\quad P_{q^{n+1}}(a) \in \{r, q \cdot r\},\ r = P_{q^n}(a)$

$q \in (\mathbb{P}_{\mathbb{N}} \setminus \{23\})\quad r = P_{q^n}(a)$

$P_{q^{n+1}}(a) = qr$

$\Rightarrow P_{q^{n+2}}(a) = q^2 r$

$m = \prod_{j=1}^{\hat{n}} q_j^{c_j} \Rightarrow P_m(a) = M(P_{q_j^{c_j}}(a),\ j \in \overline{1,n})$

Задача: Вычислить: $P_{8125}(37)$

$8125 = 25 \cdot 325 = 5^4 \cdot 13$

$P_5(37) = P_5(2)$

$\lambda(5) = \varphi(5) = 4$

$d(4) = \{1, 2, 4\}$

$2^2 \overset{2}{\not\equiv} 1$

$P_5(2) = 4$

$P_{25}(37) = P_{25}(12) = 20$

$12^2 \overset{25}{\equiv} 144 \overset{25}{\equiv} 19 \overset{25}{\equiv} -6$

$12^4 \overset{25}{\equiv} (-6)^2 \overset{25}{\equiv} 36 - 25 \overset{25}{\equiv} 11 \overset{25}{\not\equiv} 1$

$P_{625}(37) = 500$

$P_{13}(37) = P_{13}(11)$

$\lambda(13) = \varphi(13) = 12$

$d(12) = \{1, 2, 3, 4, 6, 12\}$

$11^2 \overset{13}{\equiv} 121 \overset{13}{\equiv} 4 \overset{13}{\not\equiv} 1$

$11^3 \overset{13}{\equiv} 4 \cdot (-2) \overset{13}{\equiv} 5 \overset{13}{\not\equiv} 1$

$11^4 \overset{13}{\equiv} 16 \overset{13}{\equiv} 3 \not\equiv 1$

$11^6 \overset{13}{\equiv} (11^3)^2 \overset{13}{\equiv} 25 \overset{13}{\equiv} -1$

$НОК(500, 12) = \frac{500 \cdot 12}{НОД(500, 12)} = \frac{6000}{4} = 1500$

Ответ: 1500

Задача: Вычислить наименьший первообразный корень по mod 79

$\min(PR(Z_{79}^*)) - ?$

$P_{79}(x) = \varphi(79) = 78 = 2 \cdot 3 \cdot 13$

$a \in \{2, 3, 4, 5, 6, 7, \dots\}$

$b \overset{79}{\equiv} a \quad \tilde{d}(78) \cap \mathbb{P}_{1N} = \{2, 3, 13\}$

$c_i^m \equiv b^{a/q_i}$

$c_1 \overset{79}{\equiv} b^{78/13} \overset{79}{\equiv} b^6$

$c_2 \overset{79}{\equiv} b^{78/3} \overset{79}{\equiv} b^{26}$

$c_3 \overset{79}{\equiv} b^{78/2} \overset{79}{\equiv} b^{39}$

$2^6 \overset{79}{\equiv} 64 \overset{79}{\equiv} -15$

$2^{26} \overset{79}{\equiv} (2^{12})^2 \cdot 4 \overset{79}{\equiv} (-12)^2 \cdot 4 \overset{79}{\equiv} 65 \cdot 4 \overset{79}{\equiv} -14 \cdot 4 \overset{79}{\equiv} -56 \overset{79}{\neq} 1$

$15 = 1111_2$

| | 15 | 15 | 15 | 15 |
|---|---|---|---|---|
| 2 | 15 | 45 | 26 | 67 |
| | | | | 111 |
| | | | | −12 |

$2^{39} \overset{79}{\equiv} 2^{26} \cdot 2^{12} \cdot 2 \overset{79}{\equiv} (-56) \cdot 67 \cdot 2 \overset{79}{\equiv} 23 \cdot (-12) \cdot 2 \overset{79}{\equiv} 23 \cdot (-24) \overset{79}{\equiv} -78 \overset{79}{\equiv} 1$

$23 = 10111_2$

$24 = 11000_2$

| | 23 | 23 | 0 | 0 | 0 |
|---|---|---|---|---|---|
| 2 | 23 | 69 | 138 | 118 | 78 |
| | | | 111 | 111 | 111 |
| | | | 59 | 39 | |

$3^6 \overset{79}{\equiv} 3^4 \cdot 3^2 \overset{79}{\equiv} 2 \cdot 9 \overset{79}{\equiv} 18$

$3^{26} \overset{79}{\equiv} (3^6)^2 \cdot 3^2 \overset{79}{\equiv} 18^2 \cdot 9 \overset{79}{\equiv}$